F Full Experiments

In this section, we provide additional experimental details and analyses to supplement the main results presented in the paper. We conduct experiments on various LLMs and reasoning tasks to comprehensively evaluate the effectiveness of our proposed Contrastive Denoising Chain-of-Thought (CD-CoT) method in handling noisy rationales. Specifically,

- In Appendix F.1, we describe the detailed experimental setups, including the base language models used, evaluation settings, and CoT settings.
- In Appendix F.2, we present additional experimental results on different language models, and computational costs, and introduce the Normalized Difference in Accuracy (NDA) metric for evaluating the efficacy of methods under noisy scenarios.
- In Appendix F.3, we conduct an ablation study to further justify the denoising power of the proposed CD-CoT method and highlight the importance of its two key components: contrastive denoising with rationale rephrasing and repeated reasoning with voting.
- In Appendix F.4, we explore the impact of different semantic difficulty levels of irrelevant noise on model performance and the effectiveness of CD-CoT under these scenarios.
- In Appendices F.5 and F.6, we investigate how the number of noisy thoughts per example and the number of noisy examples affect the model's reasoning performance.
- In Appendix F.7, we discuss the variants of CD-CoT, including a self-supervised variant that does not rely on manually crafted clean examples.
- In Appendix F.8, we further explore CD-CoT's robustness by introducing new datasets. Moreover, we investigate the noisy rationale problem in large-scale real-world scenarios by evaluating the impact of noisy context on model performance in multi-turn conversational QA tasks using the MT-Bench [108] dataset.
- In Appendix F.9, we provide additional qualitative results, showcasing the denoised examples generated by different robust methods across various tasks under medium-level noise.

These additional experiments and analyses aim to provide a comprehensive understanding of the noisy rationale problem, the effectiveness of the proposed CD-CoT method, and the factors influencing model performance under noisy scenarios.

F.1 Detailed Setups of the Experiments

We employ GPT-3.5-turbo-0613 [17] as our base LLM (denoted as Base) for the analyses presented in this study. In addition, we conduct evaluations on three supplementary models, including Gemini-Pro (Jan. 2024) [76], Llama2-70B [79], and Mixtral-8x7B [33]. While evaluating baseline methods on various, we consistently keep the temperature parameter τ and the top-p setting at their default value of 1, along with all other hyperparameters of models set to defaults. We conducted experiments on the first 300 questions for each task and repeated reasoning 5 times for each question.

We assume all CoT experiments with clean rationales or noisy rationales are conducted in a 3-shot setting unless specified otherwise. Furthermore, all CoT examples are constructed by randomly drawing from all available questions, except for the NoRa-Symbolic Longer task, which has predefined demonstrating and testing scopes.

F.2 Supplementary Results of the Main Experiments

Different LLMs. Fig. 7 displays the result of the GPT-3.5-Turbo model's evaluation on the NoRa Dataset. It corresponds to base model results in Tab. 3. We have also conducted comprehensive experiments on the Gemini model to evaluate various types of noise. Fig. 8 shows the full performance evaluation of Gemini on the NoRa dataset.

Computation Cost. Tab. 31 shows that the cost of CD-CoT is within an acceptable range compared with other methods.

The Normalized Difference in Accuracy (NDA) Metric. We propose a new evaluation score, Normalized Difference in Accuracy (NDA), to quantify the efficacy of $\mathcal{M}$ under the noisy scenario,

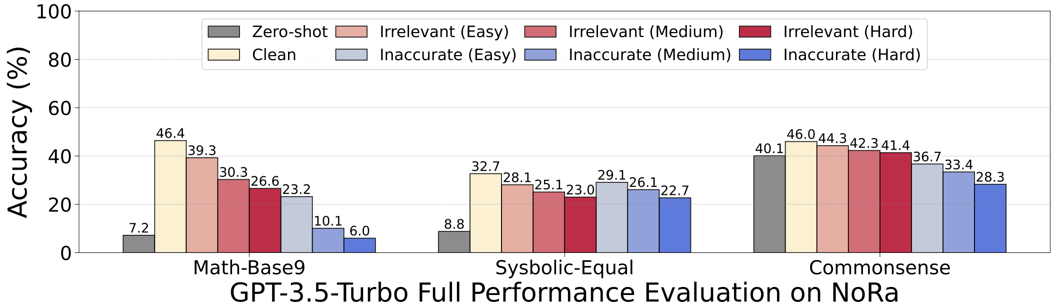

Figure 7: **GPT-3.5-Turbo** Full Performance Evaluation on the NoRa Dataset.

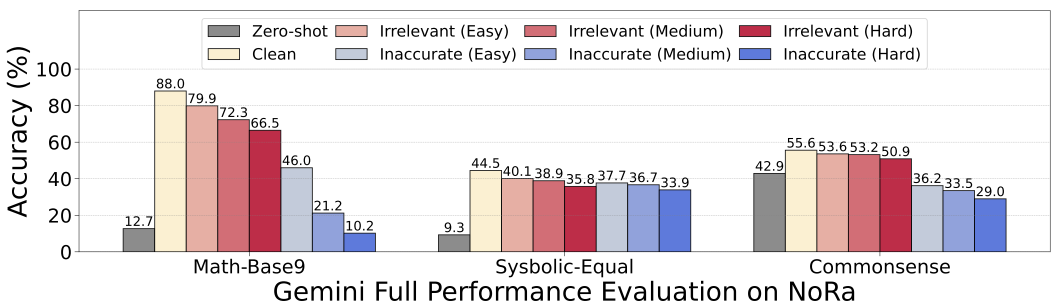

Figure 8: **Gemini** Full Performance Evaluation on the NoRa Dataset.

$$\mathrm{NDA}(\mathcal{M}, \mathcal{Q}, \mathcal{P}) = \frac{\mathrm{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{noisy}}) - \mathrm{Acc}(\mathcal{M}, \mathcal{Q}, \emptyset)}{\mathrm{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{clean}}) - \mathrm{Acc}(\mathcal{M};, \mathcal{Q}, \emptyset)}, \tag{21}$$

where $\mathrm{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{clean}})$, $\mathrm{Acc}(\mathcal{M}, \mathcal{Q}, \mathcal{P}_{\text{noisy}})$ and $\mathrm{Acc}(\mathcal{M};, \mathcal{Q}, \emptyset)$ represent the accuracy of method $\mathcal{M}$ with clean rationales, noisy rationales (irrelevant or inaccurate), and without CoT demos. Fig. 9 is the illustration of the NDA metric.

Please note that: (1) The main evaluation metric used throughout this paper is the accuracy score as introduced in Section 4; (2) The NDA metric serves as an auxiliary tool for analyzing empirical results, specifically designed to quantify how effectively a given LLM and denoising method perform under noisy scenarios.

Tab. 32 presents a comparison of the accuracy and NDA across all methods. A negative value in NDA indicates that the accuracy of noisy rationales falls below that of 0-shot. We observe that CD-CoT consistently excels in NDA, nearing 100% in most tasks and even surpassing it in certain instances.

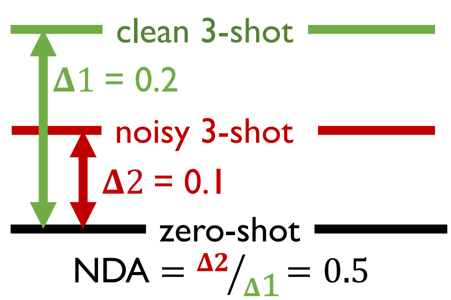

Figure 9: Illustration of the NDA metric

Standard Deviation Comparison. Tab. 33 provides a comprehensive comparison of the standard deviations (σ) across various methods on the NoRa-Math Base-9 task, using the GPT-3.5-Turbo model. The table presents the σ values for each method under different conditions, including clean rationale, irrelevant rationale (easy, medium, and hard), and inaccurate rationale (easy, medium, and hard). These results supplement the main experimental findings by offering insights into the variability and consistency of the methods' performance.

Task	Method $\mathcal{M}$	#Tokens per clean sample	#Tokens per irrelevant sample				#Tokens in inaccurate sample			
			Easy	Medium	Hard	Avg.	Easy	Medium	Hard	Avg.
Math Base-9	Base	702.9	858.2	1027.9	1209.5	1031.9	763.9	826.5	893.8	828.1
	w/ ISC [29]	1375.9	1635.8	1922.6	2224.8	1927.7	1522.1	1633.2	1735.9	1630.4
	w/ SP [89]	1664.9	1739.6	1815.9	1900.1	1818.5	1742.4	1793.2	1824.4	1786.7
	w/ SM [62]	3872.4	5029.2	5434.0	6197.6	5553.6	4226.8	4530.7	4831.1	4529.5
	w/ SD [102]	5882.6	7365.2	9453.7	11123.8	9314.2	6033.3	10819.6	11625.9	9492.9
	w/ SC [83]	2002.2	2317.4	2783.7	3362.3	2821.1	2089.0	2228.4	2416.3	2244.6
	w/ SCO [29]	3772.1	4715.5	6122.8	7250.7	6029.7	5349.2	6536.2	6942.8	6276.1
	w/ BT [81]	701.2	1706.9	1859.8	1983.5	1850.1	1603.0	1680.1	1735.0	1672.7
	w/ CC [9]	1047.7	1178.1	1315.5	1452.0	1315.2	966.2	1020.1	1075.0	1020.4
	w/ CD-CoT (ours)	5518.9	5661.9	5803.9	5933.7	5799.8	5589.2	5614.1	5647.8	5617.0
Math Base-11	Base	710.1	877.3	1057.9	1226.0	1053.7	773.1	834.4	897.3	834.9
	w/ ISC [29]	2498.2	2988.3	3531.7	4010.4	3510.1	2687.9	2872.6	3026.4	2862.3
	w/ SP [89]	2837.6	2972.3	3136.5	3199.0	3102.6	2971.2	3055.2	3119.4	3048.6
	w/ SM [62]	3945.6	5170.4	6406.1	7480.0	6352.2	4272.4	4567.2	4847.2	4562.3
	w/ SD [102]	9578.2	11613.6	13848.5	15856.9	13773.0	10094.5	10842.7	11631.1	10856.1
	w/ SC [83]	2054.5	2385.7	2872.0	3405.4	2887.7	2161.0	2289.5	2433.8	2294.8
	w/ SCO [29]	4932.4	6073.5	7403.8	8485.9	7321.1	5557.4	6554.4	6986.8	6366.2
	w/ BT [81]	1603.0	1706.9	1859.8	1983.5	1850.1	1603.0	1680.1	1735.0	1672.7
	w/ CC [9]	913.8	1064.2	1211.2	1337.0	1204.1	1127.4	1188.1	1249.9	1188.5
	w/ CD-CoT (ours)	5536.5	5681.3	5834.9	5968.1	5828.1	5647.2	5710.6	5772.9	5710.2
Symbolic Equal	Base	1623.7	1834.9	1991.0	2221.4	2015.8	1765.3	1868.8	2022.4	1885.5
	w/ ISC [29]	5246.0	5868.1	6345.2	7007.1	6406.8	5691.5	5998.2	6449.5	6046.4
	w/ SP [89]	5343.0	5539.8	5661.8	5858.0	5686.5	5526.3	5635.7	5800.3	5654.1
	w/ SM [62]	10272.9	11936.8	13156.2	14860.2	13317.7	11315.3	12029.4	13093.6	12146.1
	w/ SD [102]	6267.1	6782.5	6965.8	7004.4	6917.6	6763.1	7006.6	7135.4	6968.4
	w/ SC [83]	4089.3	4429.2	4724.0	5267.5	4806.9	4262.6	4404.0	4691.1	4452.6
	w/ SCO [29]	8937.4	10677.4	11253.8	12599.2	11510.1	10307.6	10970.7	11443.9	10907.4
	w/ BT [81]	1614.4	3058.9	3154.7	3320.2	3177.9	3011.1	3097.6	3203.4	3104.0
	w/ CC [9]	2394.7	2592.5	2734.3	2933.8	2753.5	2529.7	2625.9	2761.8	2639.1
	w/ CD-CoT (ours)	13602.0	13686.6	13609.4	13574.0	13623.3	13751.0	12818.5	13741.6	13437.0
Symbolic Longer	Base	1687.1	1826.1	1862.4	2017.9	1902.1	1802.9	1832.4	1952.3	1862.5
	w/ ISC [29]	5601.1	5957.7	6052.7	6466.0	6158.8	5932.0	6073.3	6382.8	6129.4
	w/ SP [89]	5687.9	5765.3	5815.2	5893.1	5824.5	5907.2	5911.1	6051.5	5956.6
	w/ SM [62]	10487.3	11492.5	11838.6	12922.2	12084.4	11277.9	11566.7	12346.6	11730.4
	w/ SD [102]	3087.1	3194.6	3277.5	3351.7	3274.6	3214.7	3416.0	3417.6	3349.4
	w/ SC [83]	4934.9	5153.9	5175.4	5441.8	5257.0	5146.2	5146.8	5524.5	5272.5
	w/ SCO [29]	11888.2	13232.5	13765.5	14389.6	13795.9	13255.4	12982.1	10293.9	12177.1
	w/ BT [81]	1690.1	3066.5	3091.0	3187.7	3115.1	3095.8	3154.1	3253.5	3167.8
	w/ CC [9]	2491.5	2618.7	2668.3	2801.0	2696.0	2598.5	2626.9	2737.5	2654.3
	w/ CD-CoT (ours)	14282.3	14270.6	14296.3	14286.5	14284.5	14798.8	14974.2	15393.0	15055.3
Commonsense	Base	553.2	712.0	789.5	867.6	789.7	605.9	635.6	669.9	637.1
	w/ ISC [29]	1873.2	2334.8	2559.2	2786.7	2560.2	2021.0	2110.4	2207.5	2113.0
	w/ SP [89]	4728.7	5502.0	5907.6	6308.6	5906.1	4998.5	5131.9	5265.1	5131.8
	w/ SM [62]	3532.2	4755.0	5377.2	5999.0	5377.1	3943.7	4153.7	4352.8	4150.1
	w/ SD [102]	5007.8	5882.5	6357.3	6816.1	6352.0	5285.0	5430.4	5561.9	5425.8
	w/ SC [83]	853.5	1125.7	1252.5	1382.0	1253.4	900.2	941.2	1007.3	949.6
	w/ SCO [29]	1776.8	2382.5	2688.8	2827.4	2632.9	2243.5	2372.9	2581.2	2399.2
	w/ BT [81]	553.7	1089.7	1149.5	1213.7	1151.0	984.9	1006.2	1031.1	1007.4
	w/ CC [9]	721.8	771.0	796.2	821.8	796.3	854.2	925.6	993.6	924.5
	w/ CD-CoT (ours)	3060.7	3069.9	3055.1	3098.9	3074.6	3129.5	3093.2	3090.0	3104.2

Table 31: Computation cost (#tokens) of all methods.

F.3 The Superior Performance and Denoising Effectiveness

The proposed CD-CoT method denoises noisy rationales, which leads to better reasoning performance. To be specific:

- The first two steps of CD-CoT are for explicit data denoising. First, it rephrases the noisy example by contrasting it with the clean example. Then, with the obtained rephrased examples, it selects qualified candidates by checking the validity of the rephrased answers.

- The rephrased (denoise) rationales by different robust methods are shown in Tab. 12. As can be seen, CD-CoT significantly removes noise and also ensures format alignment with the original rationale. More examples of denoising can be found in Appendix. F.9. Hence, these empirical results adequately justify the denoising power of CD-CoT.